Collinearity

Two perfectly correlated predictors

x_1: left forearm length, in centimeters

x_2: left forearm length, in meters

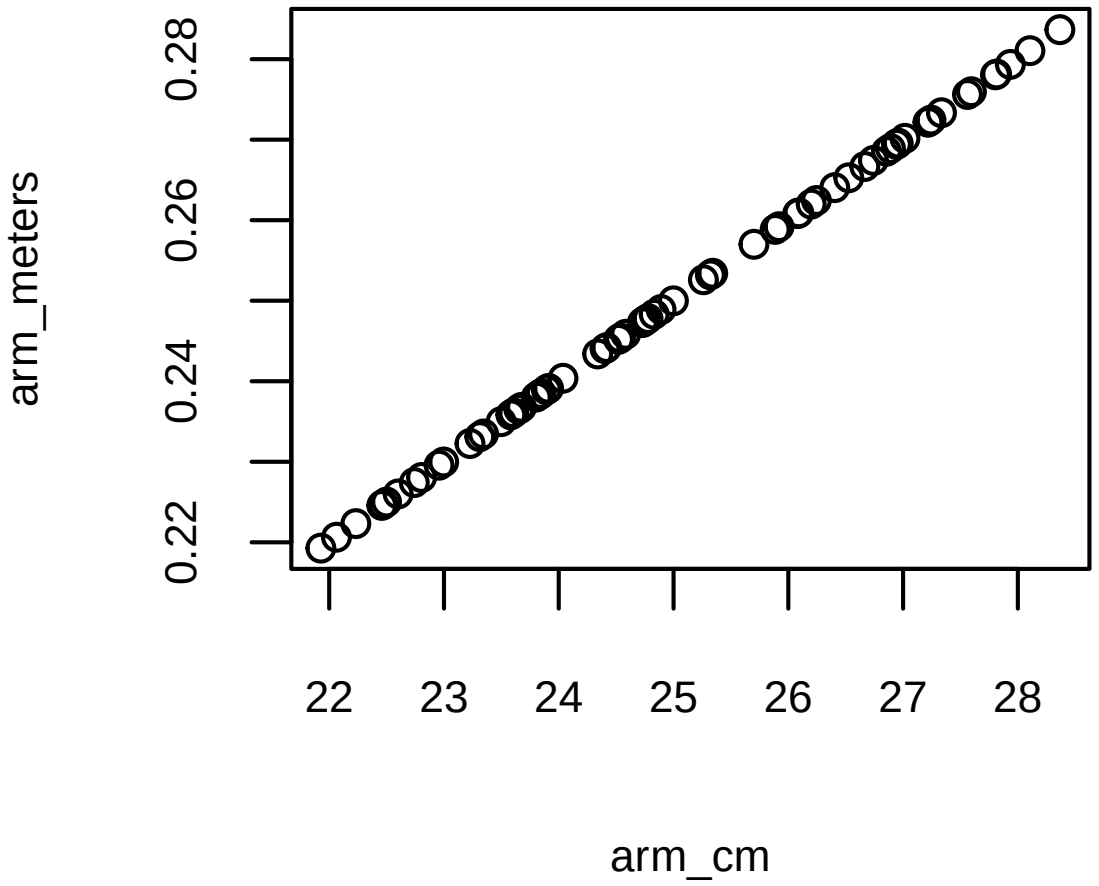
y: height, in centimeters

The arm measurement in inches is perfectly correlated with the arm measurement in centimeters:

```
cor(arm_cm, arm_meters)
```

```
## [1] 1
```

Two perfectly correlated predictors



Two perfectly correlated predictors

What happens when we try to fit a linear regression model using both predictors?

```
Call:
lm(formula = height ~ arm_cm + arm_meters)

Residuals:
   Min     1Q Median     3Q    Max
-7.389 -3.286  0.096  3.119  7.466

Coefficients: (1 not defined because of singularities)
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  42.2634     6.5731    6.43 1.65e-08 ***
arm_cm        4.8465     0.2619   18.51  < 2e-16 ***
arm_meters        NA         NA      NA       NA
---
```

Multicollinearity (exact)

Let's consider a new example with 3 predictor variables: x_1, x_2, and x_3.

We can generate pairwise scatter plots with the following command:

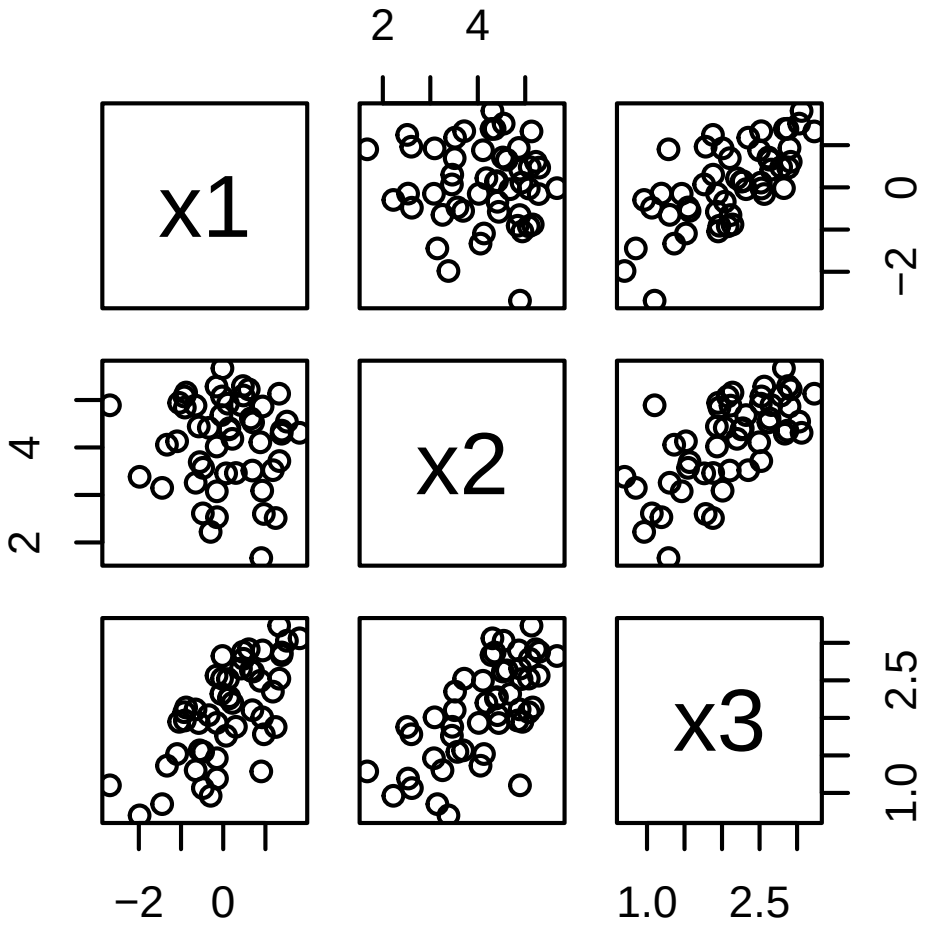
```
pairs(x)
```

(where "x" is a data frame containing those three predictors)

```
head(x)
```

```
##            x1       x2        x3
## 1 -0.5793184 4.442153 1.9314172
## 2 -1.9812952 3.382866 0.7007855
## 3 -0.1582503 5.283295 2.5625225
## 4  0.9022969 1.671243 1.2867698
## 5 -0.3406114 4.413835 2.0366119
## 6  0.1120623 4.922382 2.5172224
```

Multicollinearity (exact)



It looks like x_3 has a moderately strong correlation with x_1 and x_2. Neither is extreme- the points on the scatter plot aren't close to falling exactly on a line.

Multicollinearity (exact)

What are the actual sample correlation values?

```
cor(x)
```

```
##                x1           x2        x3
## x1  1.00000000 -0.06548474 0.6913664
## x2 -0.06548474  1.00000000 0.6756796
## x3  0.69136641  0.67567957 1.0000000
```

The correlation between x_1 and x_3 is approximately 0.7, and the correlation between x_2 and x_3 is similar. This is moderately strong, but not so strong that we would expect it to cause devastating problems for the model.

Multicollinearity (exact)

When we try to fit a linear regression model to the data, however...

```
Call:
lm(formula = y ~ ., data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-1.46962 -0.37127 -0.00831  0.35008  1.34241

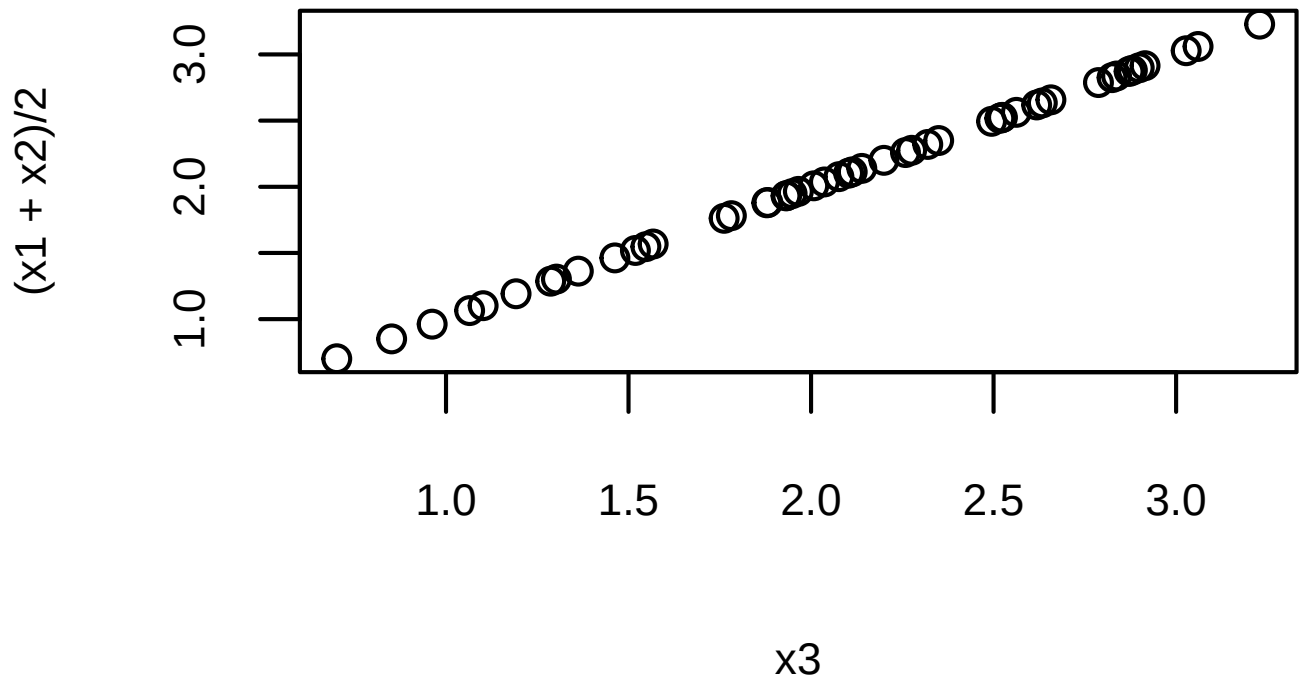
Coefficients: (1 not defined because of singularities)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.02679    0.41126   9.791 6.29e-13 ***
x1          -1.30465    0.09547 -13.666  < 2e-16 ***
x2          -1.03527    0.09741 -10.628 4.32e-14 ***
x3                NA         NA      NA       NA
---
```

The summary output shows "NA" for the coefficient for x_3. **Why**?

Multicollinearity (exact)

Because it turns out that x_3 is exactly equal to a linear combination of the other two predictors.

Specifically, x_3 is the average of x_1 and x_2 (this is simulated data).




```
cor(x3, (x1+x2)/2)
```

```
## [1] 1
```

The sample correlation of x_3 and $(x_1+x_2)/2$ is exactly equal to 1.

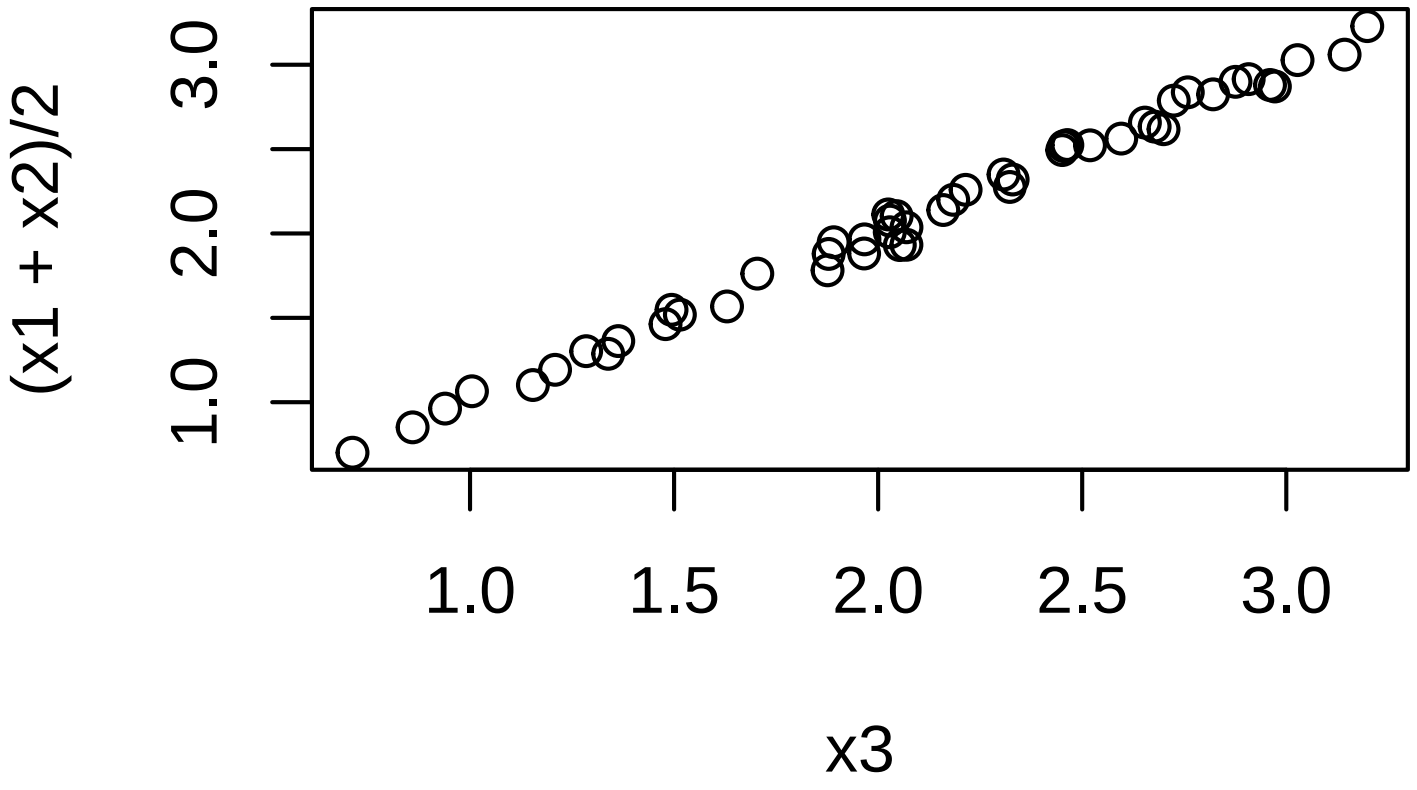
Multicollinearity (exact)

This is an issue we **can't detect by looking at pairs** of variables, because the issue is a **relationship among three variables**. Because there is perfect multicollinearity, it isn't possible to estimate the model coefficients using the method of least squares.

In general (as long as there are more data points than predictor variables), it's extremely unlikely that there will be exact multicollinearity in the observed data. But we might have variables that are very close to collinear, which we still want to detect.

Multicollinearity (approximate)

Instead, what if there is no longer exact multicollinearity, but x_3 is still highly correlated with $x_1 + x_2$?




```
cor(x3, (x1+x2)/2)
```

```
## [1] 0.9963795
```

Multicollinearity (approximate)

We can fit a linear regression model using these predictors, and the method of least squares does produce estimated coefficients:

```
Call:
lm(formula = y ~ x1 + x2 + x3)

Residuals:
     Min       1Q   Median       3Q      Max
-1.43306 -0.40510  0.01261  0.36349  1.33816

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.0438     0.4174   9.689  1.1e-12 ***
x1           -0.9787     0.8479  -1.154    0.254
x2           -0.7142     0.8355  -0.855    0.397
x3           -0.6488     1.6767  -0.387    0.701
---
```

Multicollinearity (approximate)

Check "variance inflation factors" (VIF) for the predictor variables:

```
library(car)
vif(est_model)
```

```
##        x1        x2        x3
##  77.79328  72.54703 139.62431
```

The minimum possible VIF is 1. Values larger than 5 or 10 suggest that the predictor has a large amount of correlation with other predictor variables in the model.

Multicollinearity (approximate)

What happens if we remove x_3?

```
new_model<-lm(y~x1+x2)
```

```
Call:
lm(formula = y ~ x1 + x2)

Residuals:
     Min       1Q   Median       3Q      Max
-1.46962 -0.37127 -0.00831  0.35008  1.34241

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.02679    0.41126   9.791 6.29e-13 ***
x1          -1.30465    0.09547 -13.666  < 2e-16 ***
x2          -1.03527    0.09741 -10.628 4.32e-14 ***
```

Multicollinearity (approximate)

```
sum(est_model$residuals^2)
```

```
## [1] 18.88558
```

```
sum(new_model$residuals^2)
```

```
## [1] 18.94706
```

The sum of squared residuals only increased a tiny amount when we removed x_3.

Multicollinearity (approximate)

```
summary(est_model)$adj.r.squared
```

```
## [1] 0.8482773
```

```
summary(new_model)$adj.r.squared
```

```
## [1] 0.8510221
```

The adjusted R squared increased slightly when we removed x_3.

Multicollinearity (approximate)

```
vif(new_model)
```

```
##        x1        x2
## 1.004307 1.004307
```

In the new model, the variance inflation factors for x_1 and x_2 are very small. There are no issues with correlation/collinearity among predictors in this model.